

FULL TECHNICAL FORENSIC REPORT

Digital Video Evidence Acquisition, Recovery & Compliance Documentation

Case Reference ID:	CR-2026-0089
Case Title / Action:	State of Maharashtra vs Cyber Intruder & Ors
Lead Forensic Examiner:	Insp. S. Joshi (Cyber Crime PS)
Report Generation Timestamp (UTC):	2026-09-24 07:54:44 UTC
Forensic Software Engine:	Tri-Netra DFIR Workstation (v1.0.0, Offline Forensic Workstation)
Operating Execution Environment:	Air-Gapped, Fully Offline Execution Platform
Compliance Standard Mapping:	Bharatiya Sakshya Adhiniyam (BSA 2023) §63 & ISO/IEC 27037

DOCUMENT SELF-INTEGRITY SEAL

Document Self-Integrity Checksum: Companion SHA-256 Checksum file (.sha256) generated upon document seal.
Status: Sealed and cross-referenced in hash-chained audit log immediately post-compilation.

LEGAL STANDING & STATUTORY NOTICE:

Expert-ready technical draft — requires human investigator signature; not self-certifying.

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SECTION 2: EXECUTIVE SUMMARY

This forensic report details the examination of evidentiary digital media associated with **CR-2026-0089** (State of Maharashtra vs Cyber Intruder & Ors). The primary evidence source (*Forensic Bit-Stream Image*) was analyzed under strict hardware and software write-block enforcement without modifying primary bit-stream data. A total of **2** video stream(s) were isolated across **1** independent camera channel(s), comprising **1** intact structured filesystem files and **1** heuristically recovered raw fragments. Automated AI triage cataloged **1** verified detection(s) and **1** simulated detection(s). A total of **1** manual investigator bookmark(s) were flagged. The append-only cryptographic audit chain integrity status is: **PASS**.

Total Files	Parsed Files	Carved Fragments	Verified AI	Simulated AI	Bookmarks	Audit Chain
2	1	1	1	1	1	PASS

SECTION 3: CASE & PARTY DETAILS — BSA 2023 §63 (PART A)

In accordance with the Schedule to Section 63 of Bharatiya Sakshya Adhinyam (BSA 2023), the following parameters record the identity of the digital electronic media, acquisition software, and physical sector access interfaces used to isolate electronic records.

Case Reference ID:	CR-2026-0089
Case Action / Title:	State of Maharashtra vs Cyber Intruder & Ors
Examiner / Investigator:	Insp. S. Joshi (Cyber Crime PS)
Acquisition Date / Intake:	2026-09-24T09:30:00Z
Evidence Source Media:	Forensic Bit-Stream Image
Acquisition Software:	Tri-Netra DFIR Workstation (v1.0.0)
Write-Block Verification:	Hardware/Software Write-Block Enforced (ReadOnlyHandle)
Physical Sector Reader:	Raw Block IO Crate (rust_core::raw_io)
Primary Image SHA-256:	Not yet acquired
Primary Image MD5:	Not yet acquired

MANDATORY NOTICE: Expert-ready technical draft — requires human investigator signature; not self-certifying.

SECTION 4: EVIDENCE INVENTORY

Complete itemized ledger of all extracted and recovered video evidence streams. No records are summarized or truncated. Every record is hash-verified against the source image.

File ID	Channel	Type	Size	Byte / Sector Range	Hash (SHA-256)	Timecode
CARVED_NAL_0089_FRAG.h264	Ch 1	Carved Frag	131072 B	Unallocated Cl..	a883921bf3201416..	Heuristic Recovery
CH01_20260924_100000.mp4	Ch 1	Parsed	10.00 MB	Partition 1, S..	4f53cda18c2baa0c..	2026-09-24 10:00:00

SECTION 5: ACQUISITION & INTEGRITY SUMMARY (BSA §63 PART B)

5.1 Technical Acquisition Parameters

Acquisition Method:	Hardware/Software Write-Blocked Bit-Stream Raw Read
Write-Block Status:	Enforced (ReadOnlyHandle - zero drive write transactions)
Tool Version:	Tri-Netra DFIR Workstation v1.0.0
Primary Image SHA-256:	Not yet acquired
Primary Image MD5:	Not yet acquired
Cryptographic Chain Status:	PASS (Unbroken Cryptographic Chain)

5.2 Case Methodology Description

1. Bit-Stream Acquisition & Intake: The physical evidentiary source was accessed under Hardware/Software Write-Block (ReadOnlyHandle) enforcement. Bit-stream disk image was acquired and verified using dual SHA-256 and MD5 cryptographic hashes.
2. Automated OEM Signature Detection: Scanned physical disk sectors for CCTV filesystem magic signatures; identified structure matching None specifications.
3. Structured File System Parsing: Parsed allocation tables and metadata structures, indexing 1 intact video files across 1 distinct camera channel(s). Format-preserving zero-remux copy maintained.
4. Heuristic NAL-Unit Carving: Executed unallocated sector scanner (Rust raw block IO / frame carver). Recovered 1 raw video fragment(s) from unallocated or corrupted sectors. Recovery is probabilistic and best-effort.
5. Ephemeral In-Memory Stream Decoding: In-memory hardware/software stream decoding available via memory-safe pipeline.
6. AI-Assisted Investigative Triage: Computer vision models executed in advisory mode; all outputs cataloged as investigative leads with simulation status persisted.
7. Compliance Compilation: Built Section 63 technical documentation, verified cryptographic hash chain integrity, and sealed report with companion SHA-256.

5.3 Statutory Certification Signature Blocks

STATUTORY NOTICE: This section is an expert-ready technical draft. Signature below constitutes independent review and certification by the signing expert; this software does not self-certify.

PERSON IN CHARGE OF DEVICE / MEDIA	INDEPENDENT FORENSIC EXPERT
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Signature: _____	Signature: _____
Name: _____	Name: Insp. S. Joshi (Cyber Crime PS)
Designation: _____	Designation: Forensic Video Analyst
Department / Station: _____	Department / Lab: Digital Forensics Unit
Date: _____	Date: 2026-09-24

SECTION 6: ISO/IEC 27037 ACTIVITY MAPPING

Mapping of case actions onto the four core phases of digital evidence handling specified in ISO/IEC 27037:2012. Each phase cites explicit timestamped events from the case database.

6.1 Identification Phase (1 logged events)

Evidence identification involves recognizing and cataloging potential digital media sources, OEM hardware profiles, and case parameters prior to physical seizure or analytical extraction.

- [2026-09-24 07:54:44] OEM_DETECTION

6.2 Collection Phase (2 logged events)

Collection encompasses the recovery and extraction of evidentiary data, including structured filesystem parsing, unallocated sector carving, and active channel separation.

- [2026-09-24 07:54:44] PARSE_COMPLETE
- [2026-09-24 07:54:44] CARVER_SCAN_COMPLETE

6.3 Acquisition Phase (1 logged events)

Acquisition defines the creation of a forensic bit-stream duplicate of the digital media under hardware or software write-blocking, with immediate cryptographic hash computation.

- [2026-09-24 07:54:44] ACQUISITION_VERIFIED

6.4 Preservation Phase (3 logged events)

Preservation maintains the integrity and authenticity of digital evidence via continuous append-only cryptographic hash chaining, audit verification, and tamper analysis.

- [2026-09-24 07:54:44] CASE_INTAKE
- [2026-09-24 07:54:44] TIMELINE_NORMALIZATION
- [2026-09-24 07:54:44] TAMPER_CHECK_RUN

SECTION 7: CHAIN OF CUSTODY AUDIT LOG

LIVE CRYPTOGRAPHIC CHAIN INTEGRITY: PASS

Audit Details: All 7 entries verified unbroken from Genesis to Entry #7.

ID	Timestamp (UTC)	Event Type	Entry Hash (SHA-256)
1	2026-09-24T07:54:44	CASE_INTAKE	a923a81d28291830daf1af87...
2	2026-09-24T07:54:44	ACQUISITION_VERIFIED	1d13ec3f3bdf96d38e087ea6...
3	2026-09-24T07:54:44	OEM_DETECTION	ab94ce3430ab012cba23c4a1...
4	2026-09-24T07:54:44	PARSE_COMPLETE	44f0f295838dcd3b87d9137b...
5	2026-09-24T07:54:44	CARVER_SCAN_COMPLETE	d182b33ce1a27252a4e2f111...
6	2026-09-24T07:54:44	TIMELINE_NORMALIZATION	317d0b2b3ed33116efa3f34b...
7	2026-09-24T07:54:44	TAMPER_CHECK_RUN	476b67c7673d06873eba2776...

SECTION 8: TIMELINE RECONSTRUCTION & CLOCK CALIBRATION

Per-channel clock calibration reconciles video recorder clock drift against real-world reference timestamps using On-Screen Display (OSD) timecode extraction and visual anchor change-points.

Channel	Calibration Method	Calculated Drift	Anchor Status	Verification Note
Channel 1	OSD-Only	+0.00 sec	No visual anchor found	Using on-screen timestamps only

SECTION 9: AI-ASSISTED TRIAGE FINDINGS

MANDATORY LEGAL BOUNDARY: Every AI finding in this section is strictly an investigative lead, not an identification. Computer vision outputs are non-probative leads and do not constitute definitive legal identification.

9a. Verified Findings (Weights Verified Against Registry)

Detection ID	Model Used	Class / Target	Confidence	Evidence Reference	Advisory Status
DET-001	YOLOv8-ObjectDetection	person	0.94	CH01_20260924_100000.mp4 (f#45)	VERIFIED investigative lead, not an identification

9b. Simulated Findings (Unverified / No Weights Loaded)

SIMULATION NOTICE: The following results were produced by a model running in simulated mode (no verified weights loaded) and have not been independently confirmed. They are included for completeness and must not be relied upon without separate verification.

Detection ID	Model Used	Class / Target	Confidence	Evidence Reference	Advisory Status
FACE-SIM-001	MobileNetV2-FaceDetection	Face	0.78	CH01_20260924_100000.mp4 (f#120)	SIMULATED investigative lead, not an identification

SECTION 10: INVESTIGATOR FINDINGS & MANUAL BOOKMARKS

This section contains exclusively human-authored investigative bookmarks and manual notes. No automated computer vision or AI detections are included in this section.

Bookmark ID	Evidence Reference	Investigator Note	Created By	Timestamp (UTC)
BM-001	CH01_20260924_100000.mp4 @ Frame 45 (10:01:30)	Suspect observed approaching server cabinet with unauthorized USB dongle.	Insp. S. Joshi	2026-09-24T10:45:00

SECTION 11: RECOVERED & CARVED EVIDENCE SUMMARY

RECOVERY LIMITATION CAVEAT (BLUEPRINT §5.3):
Heuristic NAL-unit recovery from unallocated or corrupted space. Frame reassembly is probabilistic and best-effort; unallocated sector recovery does not assure complete or undamaged GOP streams.

Fragment ID	Channel	Size	Sector / Byte Range	SHA-256 Hash
CARVED_NAL_0089_F RAG.h264	1	128.0 KB	Unallocated Cluster 45000-45256	a883921bf3201416...

SECTION 12: VIDEO TAMPER & ANTI-SPLICING INTEGRITY RESULTS

Statistical anti-tampering analysis was executed. Total anomalies flagged: 2.

Anomaly Type	Frame Index	Threshold / Value	Technical Description
qp_discontinuity	45	Duplicate Hash Match	
duplicate_gop	90	Duplicate Hash Match	

SECTION 13: METHODOLOGY & REPRODUCIBILITY STATEMENT

The analytical procedures implemented in this software are deterministic, air-gapped, and mathematically verifiable. The identical acquired bit-stream image processed under the identical software version will yield identical cryptographic hashes, parsed offsets, and diagnostic records. Opposing experts are invited to independently reproduce all findings using the published specifications and verifiable bit-stream media.

Subsystem Component	Engine Identifier	Implementation Architecture
Physical Bit-Stream Acquisition	engine1_acquisition (v1.0.0)	Rust raw_io block reader / ReadOnlyHandle
Proprietary Filesystem Detect	engine2_detector (v1.0.0)	Deterministic sector magic inspection
OEM Filesystem Parsers	engine3_parsers (v1.0.0)	Hikvision HIKFAT / Dahua DHFS / Heimvision FAT32
Heuristic NAL Unit Carver	engine4_carver (v1.0.0)	Unallocated NAL-unit recovery (Rust hot path)
In-Memory Stream Decoder	engine5_playback (v1.0.0)	In-memory ephemeral frame decoder (no disk write)
Timeline Drift Calibration	engine6_timeline (v1.0.0)	OSD OCR extraction + visual anchor change-point normalizer
Cryptographic Case DB	engine7_case_db (v1.0.0)	WAL SQLite + SHA-256 hash-chained audit log
AI Computer Vision Triage	engine8_ai (v1.0.0)	ONNX Runtime advisory triage with simulation flags
Compliance & Legal Export	engine10_compliance (v1.0.0)	BSA 2023 §63 & ISO/IEC 27037 technical compiler

SECTION 14: APPENDICES & TECHNICAL GLOSSARY

14.1 AI Model Registry Verification Snapshot

Model Identifier	Verification Status	Runtime Execution Mode
yolov8n.onnx	VERIFIED	Hardware Weights Loaded
scrfd_500m.onnx	VERIFIED	Hardware Weights Loaded
osnet_x0_25.onnx	VERIFIED	Hardware Weights Loaded
clip_vit_b32_text.onnx	VERIFIED	Hardware Weights Loaded
clip_vit_b32_image.onnx	VERIFIED	Hardware Weights Loaded
yolov8n_plate.onnx	VERIFIED	Hardware Weights Loaded
anpr_ocr.onnx	VERIFIED	Hardware Weights Loaded
veri776_reid.onnx	VERIFIED	Hardware Weights Loaded

14.2 Forensic Terminology Glossary

Bit-Stream Image	A bit-by-bit physical sector duplicate of storage media preserving unallocated, slack, and corrupted sectors.
Write-Block Enforcement	Physical or kernel-level software interception preventing write requests from altering evidentiary media.
Hash-Chaind Audit Log	An append-only cryptographic ledger where each event hash incorporates the SHA-256 of the prior record.
NAL Unit	Network Abstraction Layer packet containing raw H.264/H.265 compressed video payload bytes.
Carved Fragment	A video data sequence recovered heuristically from unallocated space without structured filesystem pointers.
GOP (Group of Pictures)	A sequence of successive video frames beginning with an independent Keyframe (I-frame).
OSD (On-Screen Display)	Text rendered directly onto recorded video frames displaying camera channel, date, and timecode.

14.3 Legal Disclaimer & Non-Self-Certifying Standing

LEGAL INVARIANT: Expert-ready technical draft — requires human investigator signature; not self-certifying. This software is an analytical workstation designed to assist qualified examiners. It does not self-certify. All analytical outputs, timecode calibrations, and advisory computer vision leads must be reviewed, corroborated, and signed by an authorized human expert.